BTS–IV(R/S)–06–24–2782 Reg. No. | | | | | | | | |

B

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

SE 19-206-0402 HEAT AND MASS TRANSFER OPERATIONS

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Explain different modes of heat transfer and carry out the process calculations in various geometries.
CO2: Recognise the basic methodology in designing heat exchangers and evaporators.
CO3: Estimate diffusion coefficients and convective mass transfer rates.
CO4: Select and design the most suitable equipment for absorption, distillation, liquid-liquid extractio and solid-liquid extraction.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	With a neat figure, explain the concept of velocity boundary layer for free convection of a unidirectional flow over a hot vertical flat plate.	3	L3	1	1
	(b)	What is Prandtl number? Draw the different boundary layers for Pr = 1, Pr >1 and Pr < 1 for 1D, laminar flow over a horizontal flat plate.	3	L2	1	1,2
	(c)	What are the different laws of radiation? Give the expressions of each.	3	L1	2	1
	(d)	Illustrate with a flow diagram the classification of heat exchangers.	3	L1	2	1
	(e)	Oxygen is diffusing in a mixture of oxygen-nitrogen at 1std atm, 25°C. Concentration of oxygen at planes 2 mm apart are 10 and 20 volume %, respectively. Nitrogen is non-diffusing. Calculate the flux of oxygen. Diffusivity of oxygen in nitrogen = 1.89×10^{-5} m^2/s R = 8.314 J/(mol.K)	3	L3	3	2
	(f)	Write short note on the different mass transfer theories. Give the expressions.	3	L1	3	2
	(g)	Explain drying curve with help of a diagram.	3	L2	4	1
	(h)	Explain the construction and working of a spray extraction tower, with help of a neat diagram.	3	L1	4	1

PART B

		$(4 \times 12 = 48)$				
II.	(a)	Derive the expression for heat transfer by conduction in a composite wall with parallel and series heat flow path.	6	L2	1	1,2
	(b)	Derive an expression for forced convection through a circular pipe using Buckingham's Pi theorem.	6	L2	1	1

OR

(P.T.O.)

			Marks	BL	CO	PO
III.	(a)	Explain the different regimes of boiling with help of a plot.	4	L1	1	1
	(b)	In an oil fixed furnace, the combustion products are at 1000°C and the atmospheric air is at 25°C. The convection heat transfer coefficient between the hot combustion products at the furnace wall is 10 W/m^2K and the convection heat transfer coefficient between the furnace wall and atmospheric air is 5 W/m^2K. The furnace wall is to be made with fire clay brick (k = 1.04 W/mK). Determine the maximum possible thickness of fire clay wall, if the wall temperature is hot to exceed 800°C. Draw the thermal circuit and an equivalent electrical circuit. Use 1 V potential difference to represent 100°C temperature difference and a current of 1 ampere to represent a heat flux of 1000 W/m^2.	8	L3	1	1,2
IV.	(a)	Write short note on Wilson plot method for determination of individual heat transfer coefficient.	3	L2	2	1
	(b)	A heat exchanger is used to heat 1720 kg/h of water from 293 K to 318 K with saturated steam condensing on the outside surface of brass tubes of 25 mm outer diameter and 22.5 mm inner diameter tube length is 4 m. Assuming water velocity is being constant at 1.2 m/s, determine the number of tubes required in the heat exchanger. Data: k_{brass} = 460 kJ/hmK, λ_{steel} = 2230 kJ/kg Saturation temperature of steam = 383 K Steam side coefficient = 19200 kJ/hm^2K Physical properties of water at mean fluid temperature, ρ = 995.7 kg/m^3 Cp = 4.28 kJ/kgK k = 2.54 kJ/hmK μ/ρ = 0.659×10^{-6} m^2/s.	9	L3	2	1
		OR				
V.	(a)	A single-effect vertical short-tube evaporator is used to concentrate a syrup from 10% to 40% solids at the rate of 2000 kg of feed per hour. The feed enters at 30°C and a reduced pressure of 0.33 kg/cm^2 is maintained in the vapour space. At this pressure, the liquor boils at 75°C. Saturated steam at 115°C is supplied to the steam chest. No sub-cooling of the condensate occurs. Calculate the steam requirement and the number of tubes (0.0254 m, 16 BWG), if height of the calandria is 1.5 m. The following data are given: specific heat of liquor = 0.946 kcal/kg°C latent heat of steam at 0.33 kg/cm^2 = 556.5 kcal/kg boiling point of water at this pressure = 345 K The overall heat transfer coefficient = 2150kcal/h m^2 °C i_p = 0.	10	L2	2	1
	(b)	Write the equation to find logarithmic mean temperature difference of a counter-current heat exchanger. Define the terms.	2	L1	2	1
VI.	(a)	Derive the expression for the mass transfer between two different phases. Also, explain the concept of rate controlling step.	8	L3	3	2
	(b)	What are the advantages of packed bed column over tray absorption towers?	4	L2	3	1
		OR				

BTS–IV(R/S)–06–24–2782

			Marks	BL	CO	PO
VII.	(a)	Write short note on the methods of determining packing height of a packed bed column.	4	L3	3	1
	(b)	Derive the expression of mass flux for diffusion through a stagnant gas film.	8	L3	3	2
VIII.	(a)	Explain the construction and working of Bollman extractor with a neat figure.	7	L2	4	1
	(b)	Derive the expression of operating line of in rectifying section of distillation column.	5	L3	4	2
		OR				
IX.	(a)	Derive the expression for total time of drying rate.	5	L3	4	1
	(b)	A mixture of 45 mol% n-hexane and 55 mol% n-heptane is subjected to continuous fraction in a tray column at 1 atm total pressure. The distillate contains 95% n-hexane and the residue contains 5% n-hexane. The feed is saturated vapour. The relative volatility of n-hexane in mixture is 2.36. Determine the minimum reflux ratio.	7	L4	4	1,2

Blooms's Taxonomy Level's
L1 – 15%, L2 – 35%, L3 – 44.17%, L4 – 5.83%.